

Binghong Chen

Mechanical Engineering, College of Engineering
Carnegie Mellon University, Pittsburgh, PA

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA, USA

Master of Mechanical Engineering - Research

Aug. 2025 – Present

- **GPA:** 4.0/4.0

- **Core Courses:** Modern Control Theory, Visual Learning and Recognition, Embodied AI safety

Zhejiang University

Hangzhou, China

Bachelor of Engineering in Agricultural Engineering

Sep. 2021 – Jul 2025

- **GPA:** 3.83/4.0

- **Core Courses:** Calculus, Linear Algebra, Probability and Mathematical Statistics, Principle of Automatic Control, Base of Mechanical Design, Electrical and Electronic Engineering, Fundamentals of C Programming

SELECTED PUBLICATIONS

(* indicates equal contribution, [Google Scholar](#))

- Fanghao Wang*, **Binghong Chen***, Alois Knoll, Mingchuan Zhou, et al. “STTRL-DVO: Transformer-based Reinforcement Learning for Robust Dynamic Target Tracking in Cluttered Environments” *IEEE Transactions on Robotics (TRO)*, 2026

PREPRINTS & MANUSCRIPTS UNDER REVIEW

(* indicates equal contribution)

- Yaru Niu, Zhenlong Fang, **Binghong Chen**, Hao Zhang, Chen Qiu, H. Eric Tseng, Jonathan Francis, and Ding Zhao, “Learning Versatile Humanoid Manipulation with Touch Dreaming” Under review by *International Conference on Intelligent Robots and System (IROS)*, 2026
- Qiming Wu, Xinbo Chen, **Binghong Chen**, Fanghao Wang, Sibao Hao, Mingchuan Zhou, Limin Zeng “Towards Autonomous Robot-Assisted Minimally Invasive Surgery: a Dexterous Wristed Robotic Kit (DWRK) for Surgical Tasks” Under review by *International Conference on Intelligent Robots and System (IROS)*, 2026

RESEARCH EXPERIENCE

[Safe AI Lab](#), Carnegie Mellon University

Sep 2025 – Present

Research Assistant, Supervisor: [Prof. Ding Zhao](#)

Pittsburgh, PA, USA

➤ ***EgoX: An Unified Framework for Cross-embodiment Policy Learning*** (Under preparation for CoRL)

- Aim to build a framework for cross-embodiment learning including VR-based teleoperation, data collection, model training and build an open-sourced cross-embodiment dataset for G1 humanoid, Locoman and XArm7.
- Responsible for co-leading the project.

➤ ***Learning Versatile Humanoid Manipulation with Touch Dreaming*** (Under review by IROS 2026)

- Develop a whole-body humanoid manipulation system that combines VR teleoperation with an RL-based whole-body controller. Introduce Humanoid Transformer with Touch Dreaming (HTD), which includes future hand-joint-force prediction and EMA-supervised tactile-latent prediction for contact-aware representation learning. We evaluate five humanoid manipulation tasks spanning insertion, rigid-object reorientation, deformable-object handling, tool use, and bimanual loco-manipulation.
- Assisting in data collection and task design.

➤ ***Real World RL finetuning for robotic manipulation policy*** (In Progress)

- Extend Egox framework with RL in the real world to achieve reliable policy execution. Explore what a scalable and efficient design for RL finetuning is, including human corrections and DAGger-style offline RL.
- Responsible for leading project.

Dessight Biomedical Company

Research Intern

May 2025 – Jul 2025

HangZhou, China

➤ **Object Manipulation with Bimanual Wristed da Vinci Research Kit** (Under Review by IROS 2026)

- Realized the automatic operation of multiple tasks on our independently developed microsurgical robot platform with several benchmark algorithms, including ACT, Diffusion Policy, DP3. Combine high-level policy and motion planner for fast and stable response.
- Responsible for validating IL pipelines for real robot and building RL simulation environment in SOFA and Isaac Sim for deformable object manipulation.

Robotic Micronano Manipulation Lab, Zhejiang University

Research Assistant, Supervisor: [Prof. Mingchuan Zhou](#)

Mar 2024 – Apr 2025

HangZhou, China

➤ **Capturing Dynamic Target with a Magnetic Microgripper via RL** (TRO Conditionally Accepted)

- Design a magnetic microgripper capable of 2D planar movement, object capture and transportation. Proposed a Transformer-based RL framework to help microgripper navigate within a constrained environment with lots of dynamic obstacles and capture the dynamic target.
- Responsible for designing, implementing and fine tuning the network framework and designing experiments.

➤ **Bio-inspired Magnetic Microrobot** (In Progress)

- Design a magnetic microrobot inspired by the locomotion mechanism of fish capable of navigating in 3D space. The future work is about model-based RL for navigation through narrow environment.
- Responsible for the mechanical design and fabrication of the microrobot.

Adaptive Robotic Control Lab, the University of Hong Kong

Summer Intern, Supervisor: [Prof. Peng Lu](#)

Aug 2024 – Oct 2024

Hong Kong S.A.R, China

➤ **Fault-tolerant Control for Quadrotor via a Fast RL Approach** (Deprecated)

- Trained a small drone to stabilize and track trajectory with a completely fault rotor via RLtools, a C++ RL library for accelerating the training process. Succeeded in simulation but failed in reality.
- Responsible for implementation RL in simulation and deploying network on Crazyflie.

PROJECTS

Student Agricultural Robotics Competitions | *Mechanic Design, Embedded System*

Sep 2023 – Dec 2023

- Deploy Yolo on Jeston Nano for object identification. Build PID controller for navigation on STM32 or Arduino.
- Won first place award in 2024 ASABE Student Robotics Competition about trimming leaves, Anaheim, USA and the second prize in China agricultural robot competition about cutting off strawberries, Wuhan, China.
- Responsible for mechanical design of the end-effectors and building controller for navigation and specific motion.

LCM-LoRA Distillation for Multi-View Diffusion | *Generative Model*

Nov 2025 – Dec 2025

- Try LCM-LoRA distillation for SPAD to generate multi-view images of an object quickly.
- Responsible to implement DMD2 for SPAD.

TEACHING EXPERIENCE

24351 Dynamics | *Course Assistant, Carnegie Mellon University*

Jan 2026 – Apr 2026

TECHNICAL SKILLS

Programming Skills: Python, C/C++

Specific Skills: ROS, Matlab, STM32, Solidworks, Arduino

Tools: Pytorch, Numpy, Jax, Flax, Isaac Sim, XML etc.

Hobbies: Long Distance Running, Fitness, Piano, Cooking, Table Tennis